

The depth of a lake (in meters) at coordinates (x_1, x_2) is given by the function:

$$f(x_1, x_2) = 400 - 3x_1^2x_2^2,$$

where the coordinate system is also in meters. If a swimmer is located in the middle of the lake, at coordinates $(1, -2)$, determine the direction $\mathbf{d}$ that she needs to swim in order to make the depth increase as fast as possible. Provide $\mathbf{d}$ such that its norm is 1. If she swims a distance of one meter in that direction, what is the depth of the lake at her new position? What if she swims a distance of eight meters?